CS 529 AI Engineering

For this project, you will design and develop a Multi-Agent System using the OpenAI Agents Framework.

Week 1

Day 1 — Project Planning & Team Discussion

• Meet with your team.

• Finalize the project title and write a brief abstract

• Outline key goals, expected features, and agent roles.

Day 2 — Agent Development

• Design and build the core agents required for your project

• Define agent responsibilities, prompting instructions, inputs/outputs, guardrails and interactions

Day 3 — Implement Agent Patterns

• Integrate appropriate agent patterns relevant to your system

• Improve collaboration, coordination, and task execution among agents

Week 2

Days 1–2 Add Advanced Components

• Implement **Memory** (short-term and/or long-term as needed)

• Add **Knowledge Integration** using:

o Vector Store

o RAG (Retrieval-Augmented Generation)

• Add a **database** if required for your use case

• Add Gradio UI Design

• Ensure proper **context engineering** across agents and tasks